

C.1

Simple harmonic motion (SHM)

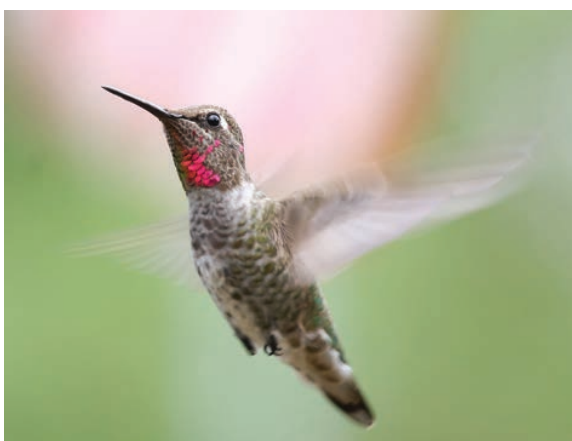
Guiding questions

- What makes the harmonic oscillator model applicable to a wide range of physical phenomena?
- Why must the defining equation of simple harmonic motion take the form it does?
- How can the energy and motion of an oscillation be analysed both graphically and algebraically?

Oscillations

◆ **Oscillation** Repetitive motion about a fixed point.

An **oscillation** is a regularly repeated backwards-and-forwards movement about the same central point, and along the same path.



■ **Figure C1.1** Oscillations of a humming bird's wings

The importance of the study of oscillations should be apparent from the very wide range of examples, both in physics and more generally. A few scientific examples:

- oscillations of the planets around the sun
- oscillations of the human heart
- oscillations of clocks (mechanical and electronic)
- oscillations of engines and motors
- oscillations of atoms within molecules
- oscillations within musical instruments
- oscillations producing human speech and within the eardrum
- oscillations of waves on water
- oscillations of light and other electromagnetic waves
- oscillations of tides on the ocean
- oscillations of electric currents.

◆ **Simple harmonic motion (SHM)** An idealized oscillation that maintains a constant amplitude and frequency.

◆ **Isochronous** Describing events that take equal times.

Simple harmonic motion (SHM) is a simplified theoretical model representing oscillations. It is the starting point for the study of all oscillations.

In perfect SHM the oscillations always take the same time and there are no resistive forces, so that they continue oscillating indefinitely with no loss of energy and constant amplitude.

This time-keeping property is described as being **isochronous**.

● Nature of science: Models

A 'model' in science means a simplified representation of a more complex situation. A model may take many forms, for example: a description in words, a drawing, a theory, an equation, a 3-D construction, a computer program or simulation, and so on.

SHM is a simple model (of oscillations) in a complex world. As the list above suggests, we are surrounded by oscillations, but few, if any, are perfect simple harmonic oscillators. Real oscillators are complex and various. To understand them, we need to first make sure we understand simplified versions.

Such simplifying models are found throughout physics and they are powerful and very useful. They should not be dismissed because of their basic assumptions.

Terms used to describe oscillations and SHM

SYLLABUS CONTENT

- ▶ A particle undergoing simple harmonic motion can be described using time period, T , frequency, f , angular frequency, ω , amplitude, equilibrium position, and displacement.
- ▶ The time period in terms of frequency of oscillation and angular frequency as given by:

$$T = \frac{1}{f} = \frac{2\pi}{\omega}$$

Although we commonly talk about oscillating masses or oscillating objects, a discussion of perfect SHM often refers to a point mass, or a 'particle'.

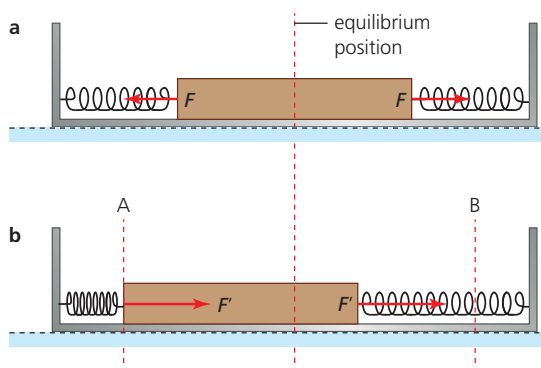
Until a mass is displaced by a resultant force, it will remain in its **equilibrium position**, where the resultant force is zero.

If a mass is then displaced, oscillations may occur if there is always a resultant *restoring force* pulling, and/or pushing, it back towards its equilibrium position. The mass will gain kinetic energy as it moves back where it came from. It will then pass through the equilibrium position

(because of its inertia), so that the displacement is then in the opposite direction. Kinetic energy is transferred to some form of potential energy in the system, and when the kinetic energy has reduced to zero, the mass will stop and then reverse its motion. And so on. The motion of a mass between two identical springs on a friction-free surface is a good visualization of this. See Figure C1.2.

At all times the springs are stretched, but in part **a** of the diagram, the forces from the springs on the mass are equal and opposite. In part **b**, there is a resultant force to the right on the mass from the springs. When released, the mass will accelerate, reach its greatest speed in the centre, and then decelerate until it stops at B. And so on.

◆ **Equilibrium position**
Position in which there is no resultant force acting on an object.



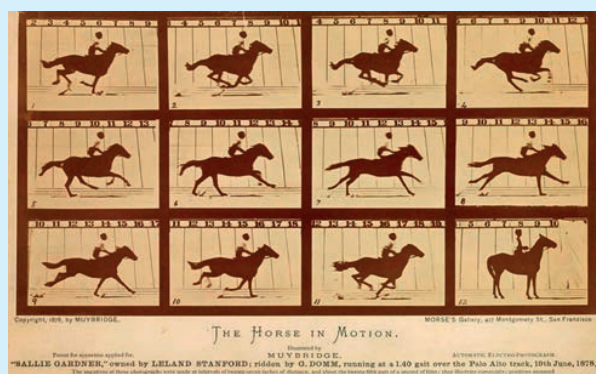
■ **Figure C1.2** Oscillations of a mass between two springs

Tool 2: Technology

Carry out image analysis and video analysis of motion

Making observations and measurements on fast-moving objects provides many technical difficulties. In the 1870s and 1880s, Eadweard Muybridge was the first to try to analyse motion (most notably of horses) by using quickly taken photographs. See Figure C1.3.

The times between each photograph were known and the position of the horse could be judged from the lines in the background. From this information, the average speed of the horse between each picture could be calculated (and, if relevant, the horse's acceleration). The pictures also revealed previously unconfirmed information about how a horse's legs moved.



■ **Figure C1.3** Famous photographs of a horse in motion (Eadweard Muybridge)

The same principles apply to *video analysis* using a modern video camera, or smart phone app. The motion to be analysed should be just in front of a suitable measurement grid. The video can be replayed in slow motion, or frame-by-frame.

Alternatively, a software program can be used which enables the position of a videoed object to be tracked and measured.

Oscillations similar to that seen in Figure C1.2 can be analysed using a suitable video camera or smart phone. A scale calibrated in millimetres can be placed behind the oscillating mass and a few seconds of action recorded while the mass is oscillating. Replaying the motion in slow motion, or frame-by-frame, can provide information on times and displacements.

Common mistake

The amplitude of an oscillation is *not* the distance between its extreme positions (which equals twice the amplitude). A single movement between the extremes is *not* a complete oscillation (it is half an oscillation).

The displacement, x , of an oscillator is its distance from the equilibrium position in a specified direction. Displacement is a vector quantity. (This term should be familiar from Topic A.1.) The displacement varies continuously, both in size and direction, during an oscillation.

The **amplitude**, x_0 , of an oscillation is its maximum displacement while oscillating.

In the idealized example (SHM), where there is no energy dissipation, the amplitude will remain constant. More realistically, each oscillation will have an amplitude which is less than the one before (assuming there is no driving force).

One *oscillation* is completed every time that an oscillating mass returns to a certain position, moving in the same direction.

A complete oscillation is sometimes called a **cycle**. An object which oscillates is called an **oscillator**.

The **time period**, T , of an oscillation is the time taken for one complete oscillation. Unit: s.

The **frequency**, f , of the motion is the number of oscillations in unit time (per second). Unit: hertz, Hz. A frequency of 1 Hz means that there is one oscillation per second.



$$\text{frequency, } T = \frac{1}{f}$$

◆ **Amplitude** Maximum displacement of an oscillation (or wave).

◆ **Cycle (oscillation)** One complete oscillation.

◆ **Oscillator** Something which oscillates.

◆ **Time period, T** Time taken for one complete oscillation.

◆ **Frequency, f** The number of oscillations per unit time, (usually per second). $f = \frac{1}{T}$ (SI unit: hertz, Hz).

The time period and frequency of an oscillation provide exactly the same information. Typically, we prefer to use the one which is greater than one.

The time period and frequency of most practical oscillations remain constant (they are *isochronous*), even when the amplitude reduces because of energy dissipation.

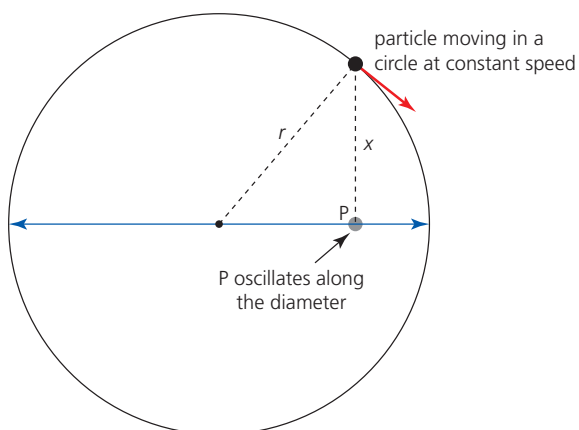
In physics we often deal with high frequencies, so the following units are in common use: kHz (10^3 Hz), MHz (10^6 Hz), GHz (10^9 Hz).

Connection between SHM and circular motion

Sometimes we may describe circular motion as an oscillation, and the last two terms described above (frequency and time period) should already be familiar because they were introduced in the circular motion sub-topic in A.2.

There is a close connection between oscillations and circular motion. Indeed, viewed from the side, motion in a circle has exactly the same pattern of movement as a simple oscillation.

Figure C1.4 shows a particle moving in a circle at constant speed. Point P is the projection of the particle's position onto the diameter of the circle.



■ **Figure C1.4** Comparing SHM to motion in a circle

◆ **Analogy** Applying knowledge of one subject to another because of some similarities.

◆ **Angular frequency, ω**
Similar to angular velocity, but used to represent the frequency of an oscillation in rad s^{-1} (because of the mathematical similarities between uniform circular motion and simple harmonic oscillations.)
 $\omega = 2\pi f$



$$\text{angular frequency, } \omega = \frac{2\pi}{T} = 2\pi f$$



Remember that, although the terms period, frequency and angular frequency are all used to describe oscillations, they are just different ways of representing exactly the same information.



Knowledge and the knower

- How do we acquire knowledge?
- How do our expectations and assumptions have an impact on how we perceive things?

Analogies, correlations and causal relationships

An *analogy* is made when we compare an understanding, process or phenomenon in one area of knowledge to another seemingly unrelated area, and we see similarities. This might enable us to understand some deeper, underlying process

that causes both the phenomena – or maybe an analogy just makes it easier to understand what is going on.

If an analogy proves to be useful, but has little, or no other validity, does that justify its use, and does it increase our knowledge of the system to which it applied? What is the difference between a correlation, a causal link and an analogy?

Consider, for example, applying the mathematics of oscillations to variations in animal populations, or to economic cycles.

WORKED EXAMPLE C1.1

Consider the oscillator shown in Figure C1.2. The oscillating mass has a length of 12 cm and the distance between A and B is 18 cm.

- Calculate the amplitude of the oscillation.
- Determine the displacement of the oscillator when its end is in position B.
- State when the mass has its greatest kinetic energy.
- If the period of the oscillator was 1.5 s, calculate its
 - frequency
 - angular frequency.

Answer

- $\frac{(18-12)}{2} = 3 \text{ cm}$
- 3 cm to the right
- When it passes through its equilibrium position.
- $f = \frac{1}{T} = \frac{1.0}{1.5} = 0.67 \text{ Hz}$
 - $\omega = 2\pi f = 2 \times \pi \times 0.67 = 4.2 \text{ rad s}^{-1}$

- 1 The central processing unit of a lap-top computer operates at 3.2×10^9 cycles per second.
 - a Express this frequency in
 - i megahertz
 - ii gigahertz.
 - b Calculate the time period of each cycle.
- 2 When a guitar string was plucked (once) it oscillated with a frequency of 196 Hz.
 - a Determine the *angular* frequency of this oscillation.
 - b Suggest how you would expect the frequency and amplitude to change in the next few seconds. Explain your answer
- 3 a Show that the angular frequency of the Earth spinning on its axis is approximately $7 \times 10^{-5} \text{ rad s}^{-1}$.
 - b Determine the total angle (rad) through which it will rotate in 100 hours.
- 4 A car engine was measured to have 3755 rpm (revolutions per minute). Calculate its angular frequency.
- 5 Using slow-motion video replay, the angular frequency of the oscillations of a humming bird's wings was found to be 272 rad s^{-1} . Determine how many times it beat its wings in one minute.

Two commonly investigated oscillators

SYLLABUS CONTENT

- The time period of a mass–spring system as given by:

$$T = 2\pi \sqrt{\frac{m}{k}}$$

- The time period of a simple pendulum as given by:

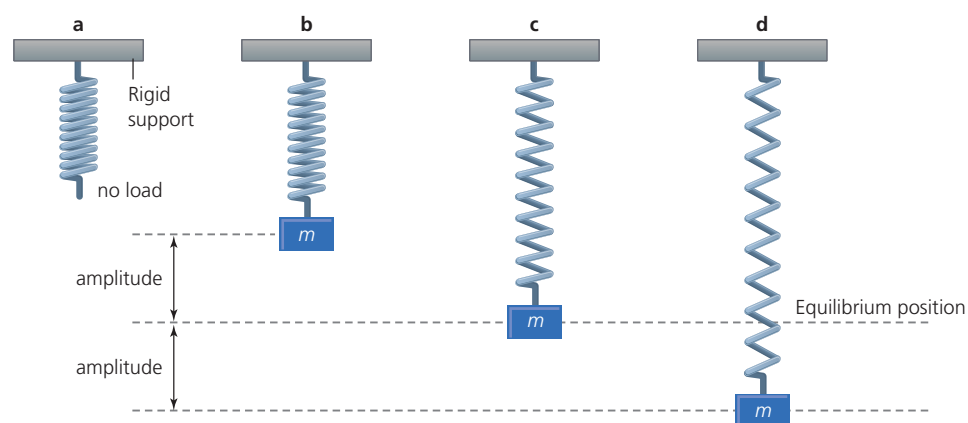
$$T = 2\pi \sqrt{\frac{l}{g}}$$

A mass–spring system and a simple pendulum are important for a number of good reasons. These include:

- The proportional relationships between force and displacement are easily understood.
- Their periods of oscillation have a convenient time for measurement in a school laboratory.
- They are good approximations to simple harmonic motion.
- Energy is dissipated slowly, so that the oscillations continue for a long enough time that allows for accurate measurements to be completed.
- They act as starting points for understanding many similar, but more complex, oscillators.

■ Mass-spring system

We have already briefly discussed a mass oscillating horizontally between two springs (Figure C1.2). A more common arrangement is shown in Figure C1.5. Of course, in this arrangement, the force of gravity (the weight of the mass) also acts vertically, but it does not affect the results because it is constant. The system would have the same time period if it was moved to a location where the gravitational field strength was different.

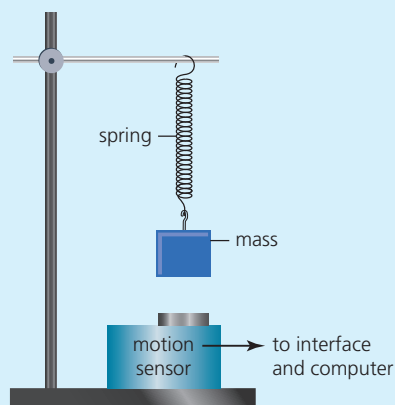


■ **Figure C1.5** Different positions of a mass, m , oscillating on a spring

Tool 2: Technology

Use sensors

Position sensors are useful in many aspects of the study of motion, including mechanical oscillations. Figure C1.6 shows how digital data can be collected, which will be processed later by the computer.



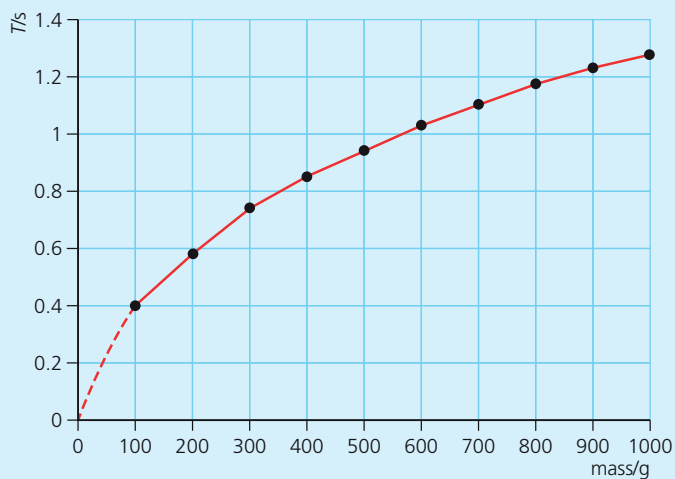
■ **Figure C1.6** Using a sensor to investigate oscillations of a mass on a spring.

We will assume that the spring is never overstretched and that it obeys Hooke's law (see Topic A.2):

$$F_H = -kx$$

Or, in words: the size of the restoring force acting on the mass = spring constant $\times$ displacement from its equilibrium position. The spring constant is a measure of the spring's stiffness (= force / deformation).

Laboratory investigations of the time periods of a mass on a spring are straightforward, especially those involving using different masses on the same spring. For typical results see Figure C1.7. Different springs, or combinations of springs, used with the same mass, can be used to investigate the effect of the spring constant, k , on the period.



■ **Figure C1.7** Variation of time period, T , with mass, m , on a spring

Tool 3: Mathematics

Linearize graphs

You may have learnt how to linearize graphs in Topic B.1. The graph seen in Figure C1.6 appears that it might represent the relationship $T^2 \propto m$. Check this by using information from the graph to draw a T^2 - m graph (uncertainties in T are ± 0.05 s). It should produce a straight line through the origin. Use the graph and the relationship shown below to determine a value for the spring constant, k .

The exact relationship for the SHM of a mass on a spring is as follows:

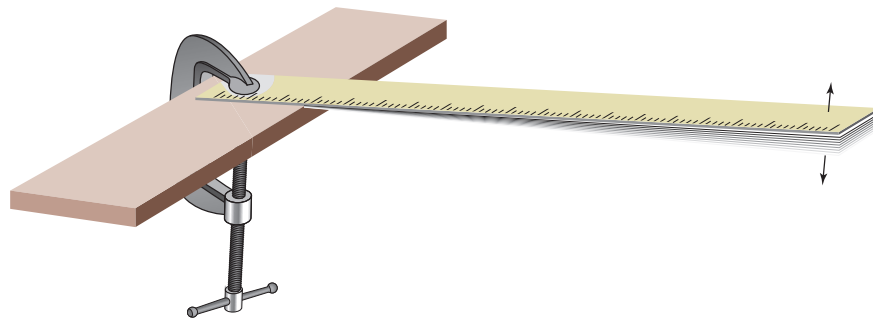
Time period of a mass-spring system:



$$T = 2\pi \sqrt{\frac{m}{k}}$$

Since $\omega = \frac{2\pi}{T}$, this equation can also be written as: $\omega^2 = \frac{k}{m}$.

This equation applies perfectly only to a point mass acted upon by a separate simple spring system that has a well-defined stiffness (k) and no (significant) mass. There is an extremely large number of other oscillating mechanical systems that have similarities to this simple model but are much more complex. For example, oscillations in buildings and bridges. Figure C1.8 shows a simpler example, in which the mass is spread uniformly along the oscillating system, a ruler.



■ Figure C1.8 Oscillating ruler

Inquiry 3: Concluding and evaluating

Concluding

A student wants to investigate an oscillating ruler, similar to that seen in Figure C1.8, but she varied the mass of the ruler by taping various masses near to its free end. She predicted that the time period of the oscillator could be determined from the same equation as for a mass oscillating vertically on the end of a spring. Table C1.1 summarizes her raw data. Assume that the uncertainties are low.

■ Table C1.1 Results of vibrating ruler experiment

Mass on end of blade / g	Time period / s
0	too quick to measure
40	0.55
80	0.76
120	0.91
160	1.04
200	1.15

Process the results and reach a conclusion. State whether the student predicted correctly. If not, suggest a possible reason.

WORKED EXAMPLE C1.2

A 200 g mass was placed on the end of a long spring and increased its length by 5.4 cm.

- Determine the spring constant, k , of the spring.
- If the mass is displaced a small distance from its equilibrium position and undergoes SHM, calculate the frequency of oscillations.
- Suggest a possible reason why oscillations with greater amplitude may not be simple harmonic.

Answer

a $F_H = -kx$

$$0.200 \times 9.8 = -k \times (5.4 \times 10^{-2})$$

$k = -36 \text{ N m}^{-1}$ (the negative sign shows that force and displacement are in opposite directions)

b $T = 2\pi\sqrt{\frac{m}{k}} = 2 \times \pi \times \sqrt{\frac{0.200}{36}} = 0.47 \text{ s}$

$$f = \frac{1}{T} = \frac{1}{0.47} = 2.1 \text{ Hz}$$

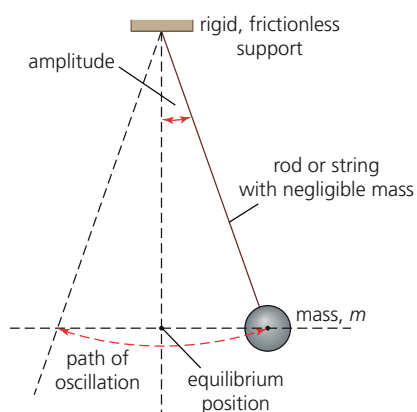
- c** The spring may become overstretched, so that Hooke's law is no longer applicable.

Simple pendulum

◆ **Pendulum** A weight, which is suspended below a pivot, which is able to swing from side to side. The weight is sometimes called the **pendulum bob**. The concept of a **simple pendulum** is a point mass on the end of an inextensible string.

There are many different designs of **pendulum**, all of which involve a mass swinging from side to side under the effects of gravity. We use the term '**simple pendulum**' to describe the simplest possible model of a pendulum: a spherical mass swinging on a rod, or string (both of which have negligible mass), from a rigid, frictionless support. See Figure C1.9. The mass is often called a **pendulum 'bob'**.

Because the mass is spherical and the rod / string has negligible mass, all of the mass of the pendulum can be assumed to be acting as a point mass at the centre of the sphere. The displacement of the motion can be considered to be the angle between the rod / string and the vertical. The restoring force is provided by gravity: consider Figure C1.10.

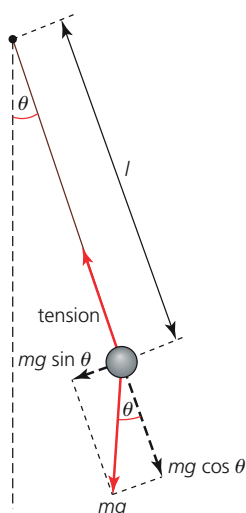


■ **Figure C1.9** Simple pendulum

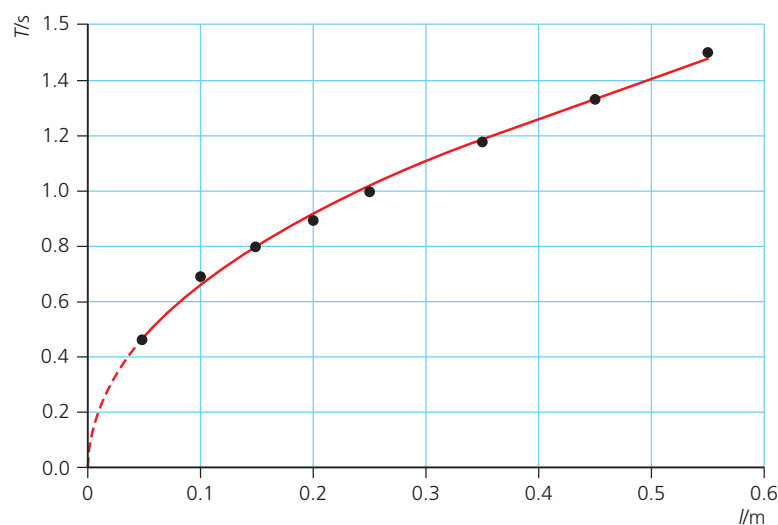
The weight, mg , of the pendulum is conveniently resolved into two perpendicular components: a force $mg \sin \theta$ provides the restoring force bringing the pendulum back towards its equilibrium position; a force $mg \cos \theta$ keeps the rod / string in tension.

An alternative way of looking at this situation: the pendulum has only two forces acting on it: tension in the rod / string and weight, mg . The resultant of these is the restoring force, $mg \sin \theta$. Practical laboratory investigations of a simple pendulum will confirm that, *for small amplitudes*, the time period, T , depends only its length, l (measured from its point of support to its centre of mass). See Figure C1.11.

Period is *not* affected by its mass. This is because doubling the mass, for example, will also result in doubling the restoring force.



■ **Figure C1.10** Components of the weight of the pendulum



■ **Figure C1.11** Results of simple pendulum investigation

The exact relationship for the SHM of a simple pendulum (for small amplitudes) is as follows.

Time period of a simple pendulum:



$$T = 2\pi \sqrt{\frac{l}{g}}$$

Top tip!

The data of Figure C1.11 can be linearized. A graph of T^2-l should be a straight line through the origin. A simple pendulum experiment can be used to determine a value for the gravitational field strength, g . The graph of T^2-l will have a gradient of: $4\pi^2/g$

Tool 3: Mathematics

Use approximation and estimation

For the motion of a simple pendulum to be a good example of SHM, we require that the restoring force ($mg \sin \theta$) is proportional to the displacement (θ). For very small angles, values of $\sin \theta$, $\tan \theta$ and θ (in rad) are almost identical, so that this condition is satisfied. But, as the angle increases, the difference gets greater. Determine the minimum angle for which there is at least a 1% difference between $\sin \theta$, $\tan \theta$ and θ (in rad).



■ **Figure C1.12** Mount Nevado Huascarán in Peru is reported to have the lowest gravitational field strength on Earth; its summit (6768 m) is one of the farthest points on the surface of the Earth from Earth's centre

ATL C1A: Research skills, thinking skills

Use search engines effectively; providing a reasoned argument to support conclusions

The equation highlighted above shows that the period of a simple pendulum would also change if its value was checked in a location where the gravitational field strength had a different value. Using your own research, investigate the variations of the strength of the gravitational field around the Earth. Then state and explain whether experiments in school laboratories would be able to measure those differences.

WORKED EXAMPLE C1.3

Determine a value for the time period of a simple pendulum of length 85.6 cm at a location where the gravitational field strength is

- a 9.81 N kg^{-1}
- b 1.63 N kg^{-1} (on the Moon's surface).
- c This equation predicts that the ratio of the time periods at two different locations is:

$$\sqrt{\frac{g_1}{g_2}}$$

Do your answers to **a** and **b** confirm that?

Answer

$$\text{a } T = 2\pi\sqrt{\frac{l}{g}} = 2 \times \pi \times \sqrt{\frac{0.856}{9.81}} = 1.86 \text{ s}$$

$$\text{b } 2 \times \pi \times \sqrt{\frac{0.856}{1.63}} = 4.55 \text{ s}$$

$$\text{c } \frac{T_M}{T_E} = \frac{4.55}{1.86} = 2.45$$

$$\sqrt{\frac{g_E}{g_M}} = \sqrt{\frac{9.81}{1.63}} = 2.45$$

Yes, the same ratio is confirmed.



Nature of science: Measurements

Technological developments in the measurement of time

Famously, Galileo was the first to discover the constant time periods of pendulums, but it was not until about sixty years later, in 1656, that the first pendulum clock was invented by the young Dutch scientist and inventor, Christiaan Huygens. Huygens is widely considered to be one of the greatest scientists and astronomers of all time. By the end of the eighteenth century, the best pendulum clocks could be made with an inaccuracy of less than one second a day. They were used throughout the world as the most accurate timekeepers for more than 250 years.

Accurate timekeepers were also essential for navigation on long journeys by ship. East–west distances (longitude) could be determined from observation of the stars or planets, combined with knowledge of the exact time of observation.



Figure C1.13
Christiaan Huygens



Figure C1.14 Clocks like this were found in many homes

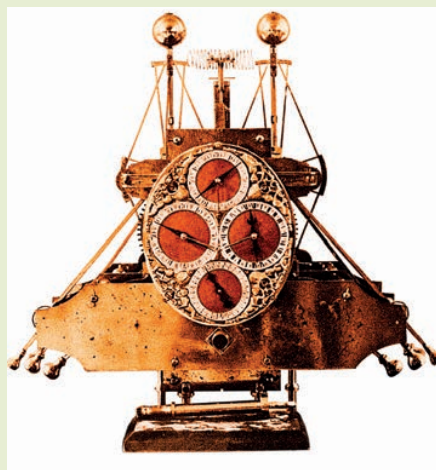


Figure C1.15 James Harrison's first marine timekeeper (1735)

However, pendulum clocks were not designed to cope with the motions of ships on the ocean! The British government offered a valuable reward to anyone who could design a clock that would remain accurate at sea, and hence 'solve the problem of longitude'. It was won by a clockmaker named James Harrison. His design included oscillating spheres on springs, rather than a pendulum.

Today we expect to know the exact time, at any time and place that we want, and we consider such precision to be normal, unworthy of comment. But, before pendulum clocks, most people were unaware of the time to the nearest hour, never mind the nearest minute, or second. (And they probably would not have understood any need for such accuracy!)

It is impossible to over-estimate the importance of the accurate measurement of time in modern life. The physics principle behind the pendulum clock is easily understood, but it took great technological skill (for that era) to manufacture the accurate clocks which had such considerable benefits on everyday life.

6 Determine what mass will have a period of exactly one second when oscillating on the end of a spring which has a spring constant of 84 N m^{-1} .

7 Calculate the angular frequency of a mass–spring system which involves a mass of 1000 g and a spring with $k = 100 \text{ N m}^{-1}$.

8 A student investigated the effect of using springs of different stiffness (k) on the periods of oscillations, T , of the same mass.
Sketch the T – k graph that should be obtained.

9 Perhaps the world’s most famous pendulum was made by Léon Foucault in Paris in 1851. The bob had a mass of 28 kg and its length was 67 m .

- a Calculate the period of this pendulum (the gravitational field strength in Paris is 9.81 N kg^{-1}).
- b Suggest reasons why it could continue to swing for a long time without any continuous energy input.
- c Use the internet to find out why this pendulum was so important.

10 Figure C1.11 shows the results of an investigation into a simple pendulum.

- a Calculate the angular frequency which describes oscillations of a pendulum of length 0.50 m .
- b Determine a value for the strength of the gravitational field from these results.

11 Figure C1.16 shows a girl starting to bounce on a trampoline. To begin with, her feet remain in contact with the rubber sheet and her movement can be considered to be SHM. The girl has a mass of 38 kg and the sheet stretched down by 33 cm when she was standing still in the middle.

- a Calculate a value for the spring constant of the trampoline.

b Determine the time period of her bounces while she remains in contact with the trampoline’s surface.



■ Figure C1.16 Girl on trampoline

12 Figure C1.17 shows a spring which is part of a car’s suspension system.



■ Figure C1.17 The suspension system of a car

- a Estimate its spring constant by considering how much the wing of a car will depress if you push down hard on it (or sit on it).
- b By considering that the wheel effectively oscillates on the spring, estimate
 - i its time period of oscillation
 - ii how far a car travelling at 12 ms^{-1} moves during one oscillation.
- c The suspension also incorporates a shock absorber (damper). Discuss why this is necessary.

LINKING QUESTIONS

- How can greenhouse gases be modelled as simple harmonic oscillators?
- What physical explanation leads to the enhanced greenhouse effect? (NOS)

These questions link to understandings in Topic B.2.

Conditions that produce SHM

SYLLABUS CONTENT

- Conditions that lead to simple harmonic motion.
- The defining equation of simple harmonic motion as given by: $a = -\omega^2 x$.

SHM will occur if the restoring force, F , is proportional to the displacement, x , but the force always acts back towards the equilibrium position:

$$F \propto -x$$

The negative sign is important here. It shows us that the force acts in the opposite direction to the displacement: an increasing force *opposes* increasing displacement.

For an oscillating mass on a spring which obeys Hooke's law ($F_H = -kx$) this condition is obviously satisfied. For a simple pendulum, the restoring force, $mg \sin \theta$, is (almost) proportional to the displacement angle, θ , if the angle is small.

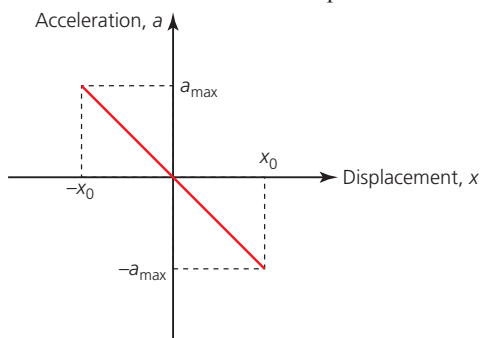
To define SHM we need to refer to the motion, rather than the force. Since $F = ma$, for a constant mass, we can write $a \propto -x$.

◆ **Simple harmonic motion (SHM)** Oscillations in which the acceleration, a , is proportional to the displacement, x , and in the opposite direction, directed back to the equilibrium position. $a \propto -x$.

Simple harmonic motion is defined as an oscillation in which the acceleration is proportional to the displacement, but in the opposite direction (always directed back towards the equilibrium position):

$$a \propto -x$$

This is represented by the graph shown in Figure C1.18. A graph of the restoring force against displacement would look similar.



■ **Figure C1.18** Acceleration–displacement graph for SHM



If, for example, the maximum displacement (amplitude) of an SHM is doubled, the restoring force and acceleration will also double. This will result in the mass taking the same time for each oscillation, because it is moving twice the distance at twice the average speed. This explains a defining feature of SHM: amplitude does not affect time period.

We can rewrite the equation above as: $a = -\text{constant} \times x$.

The constant must involve frequency because the magnitude of the acceleration is greater if the frequency is higher. The constant can be shown to be equal to ω^2 . So that the defining equation of SHM can be written as:

$$a = -\omega^2 x$$

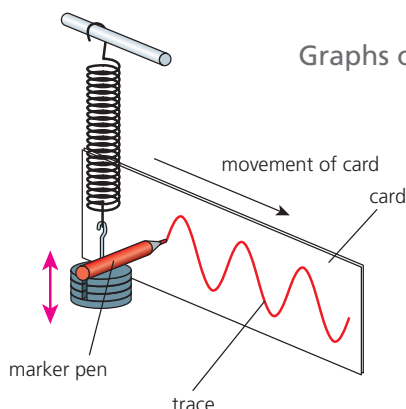
WORKED EXAMPLE C1.4

A mass oscillates horizontally between two springs with a frequency of 1.4 Hz.

- a** Calculate its angular frequency.
- b** Determine its acceleration when
 - i** its displacement is 1.0 cm to the right
 - ii** its displacement is 4.0 cm to the left
 - iii** it passes through its equilibrium position.

Answer

- a** $\omega = 2\pi f = 2\pi \times 1.4 = 8.8 \text{ rad s}^{-1}$
- b i** $a = -\omega^2 x = -(8.8)^2 \times (+0.010) = -0.77 \text{ m s}^{-2}$ (to the left)
- ii** $a = -\omega^2 x = -(8.8)^2 \times (-0.040) = +3.1 \text{ m s}^{-2}$ (to the right)
- iii** $a = -\omega^2 x = -(8.8)^2 \times 0.0 = 0.0 \text{ m s}^{-2}$



■ **Figure C1.19** Recording the motion of an oscillation

◆ Data logging

Connecting sensors to a computer with suitable software to enable physical quantities to be measured and processed digitally.

◆ **Sinusoidal** In the shape of a sine wave (usually equivalent to a cosine wave).

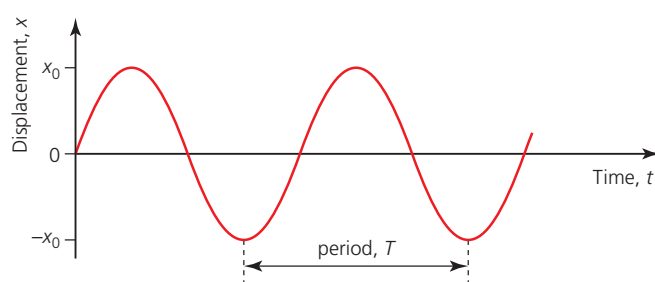
Graphs of SHM

A mass oscillating on a spring could be used with a marker pen to produce a record of the oscillation, as shown in Figure C1.19. If the card moves at a constant speed, the record (trace) produced is effectively a graph of displacement against time.

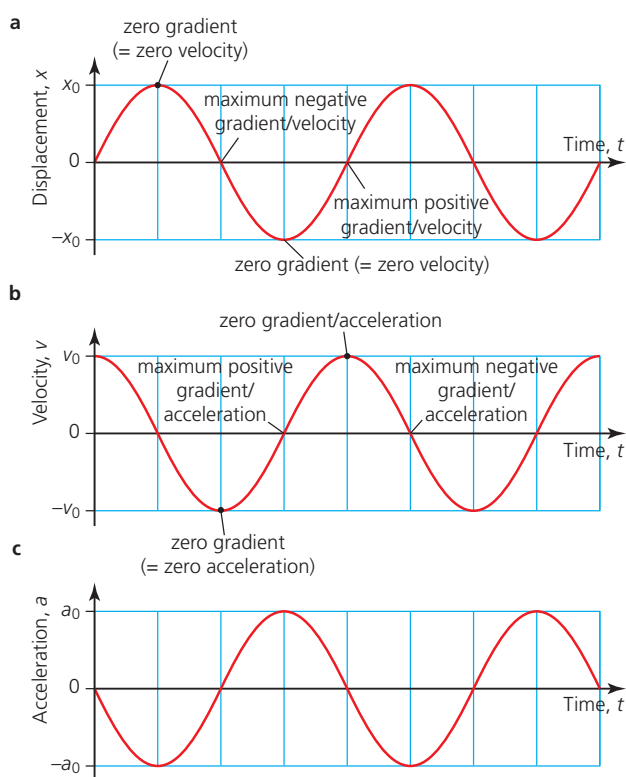
Data logging with an appropriate sensor will improve this experiment.

The waveform seen in Figure C1.19 is commonly described as being **sinusoidal** in shape.

The graph in Figure C1.20 shows the variation in displacement, x , with time, t , for the idealized model of a mass moving with perfect SHM. Here we have *chosen* that the particle has zero displacement at the start of the timing. The motion has an amplitude of x_0 . Alternatively, we might have chosen the graph to start with the particle at maximum displacement (or any other displacement).



■ **Figure C1.20** Displacement–time graph for simple harmonic motion. Timing was started when the particle had zero displacement



■ **Figure C1.21** Graphs for simple harmonic motion starting at displacement $x = 0$: **a** displacement–time **b** velocity–time **c** acceleration–time

Using knowledge from Topic A.2, we can use this graph to calculate the velocity of the oscillating mass at any particular moment by determining the gradient of the displacement–time graph at that moment:

$$\text{velocity, } v = \frac{\text{change in displacement}}{\text{change in time}} = \frac{\Delta x}{\Delta t}$$

Similarly, the acceleration at any given time can be found from the gradient of the velocity–time graph:

$$\text{acceleration, } a = \frac{\text{change in velocity}}{\text{change in time}} = \frac{\Delta v}{\Delta t}$$

Using this information, three separate but interconnected graphs of motion can be drawn and compared, as shown in Figure C1.21.

The velocity graph has its maximum value, v_0 , when the displacement, x , is zero, and the velocity is zero when the displacement is at its maximum, $\pm x_0$.

The acceleration has its maximum value when the velocity is zero and the displacement is greatest. This is to be expected, because when the displacement is greatest, the restoring force, acting in the opposite direction, is greatest.

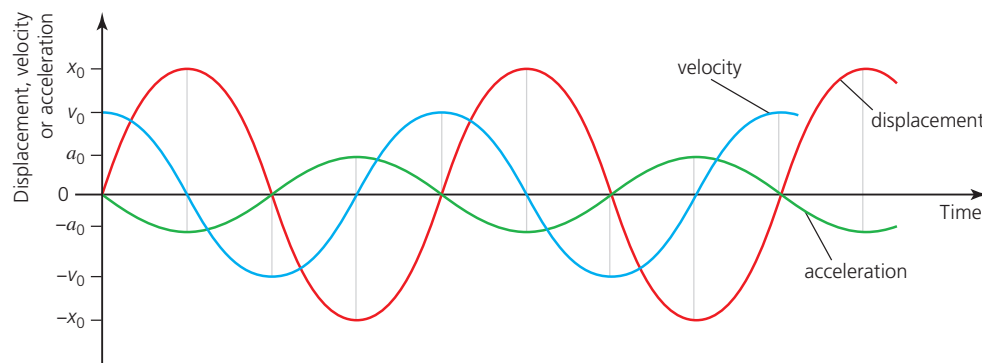
SHM graphs of displacement, velocity and acceleration are all sinusoidal in shape, but their maximum values occur at different times.

Figure C1.22 shows all three SHM graphs drawn on the same axes, so that they can be more easily compared. Note that the amplitudes of the three graphs are arbitrary; they are not interconnected and should not be compared.

LINKING QUESTION

- How can the understanding of simple harmonic motion apply to the wave model? (NOS)

This question links to understandings in Topic C.2.



■ Figure C1.22 Comparing displacement, velocity and acceleration for SHM, with timing starting at displacement $x = 0$



■ Figure C1.23 Four oscillations out of phase

Phase difference

This is a convenient point to introduce the important concept of **phase difference**. Figure C1.23 shows an everyday example. The motions of the four children all have the same frequency because all the swings are the same length, but they are each at different points in their oscillations: we say that they are *out of phase* with each other.

A phase difference occurs between two similar oscillations if they have the same frequency, but their maximum values do not occur at the same time.

◆ **Phase difference** When oscillators that have the same frequency are out of phase with each other, the difference between them is defined by the angle (usually in terms of π radians) between the oscillations. Phase differences can be between 0 and 2π radians.

The three graphs shown in Figure C1.21 all have the same frequency and sinusoidal shape, but their peaks occur at different times: there is a phase difference between the waves. This could be quantified by referring to the fraction of a time period, T , that occurs between their peaks. The first peak of the displacement graph occurs $T/4$ after the first peak of the velocity graph. The first peak of the acceleration graph occurs $3T/4$ after the first peak of the velocity graph.

But, remembering that one complete oscillation can be considered as equivalent to moving through an angle of 2π radians, phase differences are more usually quoted in terms of π .

Examples of phase differences

- One quarter of an oscillation, $\frac{T}{4} = \frac{2\pi}{4} = \frac{\pi}{2}$ radians (90°)
- One half of an oscillation, $\frac{T}{2} = \frac{2\pi}{2} = \pi$ radians (180°)
- Three quarters of an oscillation, $\frac{3T}{4} = 2\pi \times \frac{3}{4} = \frac{3\pi}{2}$ radians (270°)

(A phase difference of $\frac{3\pi}{2}$ radians is equal in magnitude to a phase difference of $\frac{\pi}{2}$ radians, but in some circumstances, we might be concerned about which peak came ‘first’.)

Referring back to Figure C1.21, can see that there is a phase difference of $\frac{\pi}{2}$ between displacement and velocity, and a phase difference of π between displacement and acceleration.